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FOOD NUTRITION CLUSTERING USING PRINCIPAL COMPONENT ANALYSIS

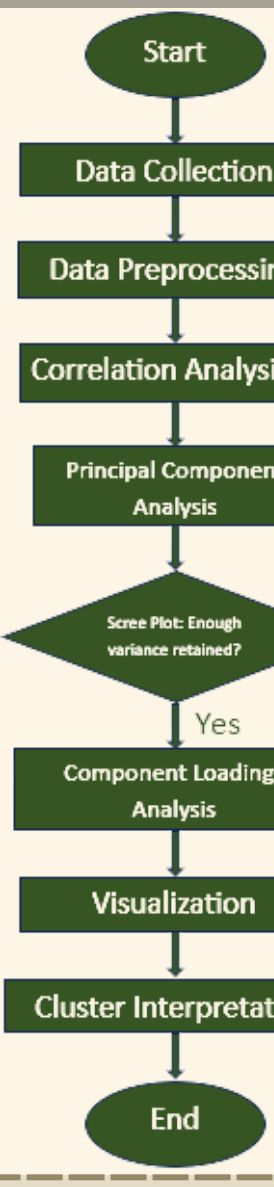
PROBLEM STATEMENT

Consumers today are overwhelmed by complex nutritional information found on food labels. With so many variables calories, fats, sugars, vitamins, and minerals make it hard to understand which foods are healthy and how they compare to each other. This project tackles the problem of information overload in food nutrition data by using Principal Component Analysis (PCA) to simplify the data and reveal hidden patterns. By clustering foods based on their nutritional profiles, we aim to help people better understand, compare, and choose foods that match their dietary goals.

DATA DESCRIPTION

This dataset was compiled from the Malaysian Food Composition Database (myfcd.moh.gov.my), containing over 30 nutritional features across various food items, including macronutrients, micronutrients, and other dietary values. The target variable in this study is the overall nutritional profile of each food item, which is analyzed using Principal Component Analysis (PCA) and clustering to group foods with similar nutrient characteristics.

METHODOLOGY



Start

Data Collection: Gather food nutrition data from myfcd.moh.gov.my and Kaggle.

Data Preprocessing: Clean data, remove irrelevant columns, handle missing values, and standardize all numeric features.

Correlation Analysis: Use a heatmap to check for multicollinearity between nutrients and justify PCA.

Principal Component Analysis: Use a heatmap to check for multicollinearity between nutrients and justify PCA.

Scree Plot: Use a scree plot to decide how many components to keep (e.g., top 2 or 3).

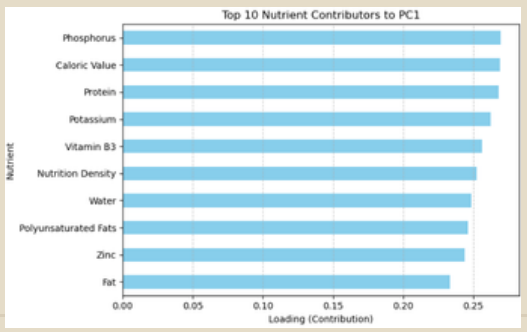
Component Loadings Analysis: Identify which nutrients contribute most to each principal component.

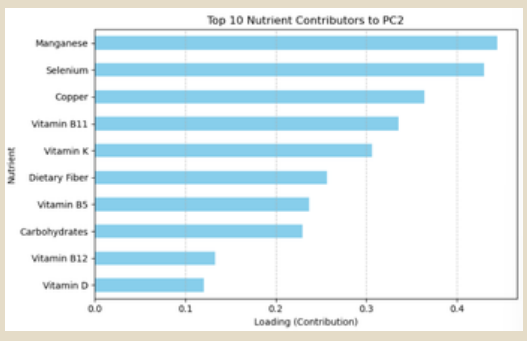
Visualization: Visualize food items and nutrient directions in reduced PCA space.

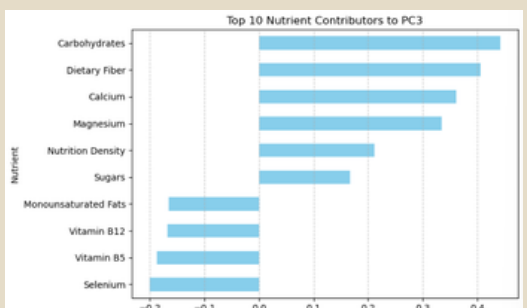
Cluster Interpretation: Analyze and interpret food groupings based on nutritional similarity.

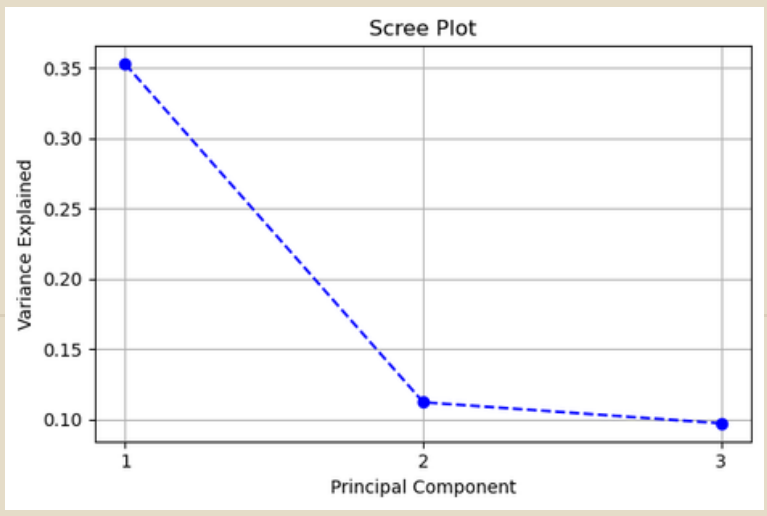
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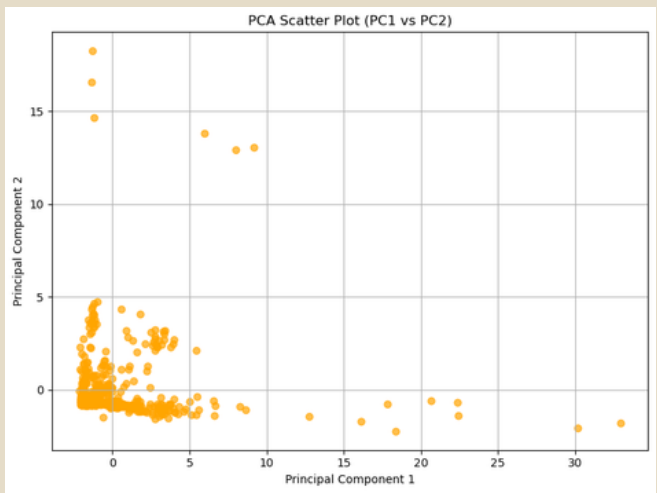
RESULTS AND MODEL EVALUATION











The first 3 principal components (PC1–PC3) selected because it explained the majority of the variance in the data. PCs beyond this point added minimal new information, based on the scree plot and cumulative variance.

PCA was used to reduce over 30 nutrition features into just 3 components: PC1, PC2, and PC3. This makes it easier to compare and group foods.

- PC1 represents foods high in energy and macronutrients like calories, protein, and fat.
- PC2 highlights micronutrient-rich foods with vitamins and minerals.
- PC3 focuses on fiber and minerals like calcium and magnesium.

The scree plot shows these 3 PCs explain most of the variance. The scatter plot (PC1 vs PC2) shows how foods group based on their nutrition profiles.

CONCLUSION

This project applied Principal Component Analysis (PCA) to reduce the complexity of a food nutrition dataset containing over 30 features. The analysis focused on the first three principal components, which captured most of the important information and allowed for clearer interpretation of the data. PC1 represented energy-dense nutrients like calories, protein, and fat. PC2 highlighted foods rich in micronutrients such as vitamins and minerals, while PC3 focused on fiber and minerals like calcium and magnesium. These components helped group similar foods and made it easier to interpret nutrition patterns and identify distinct clusters of food types. The scree plot confirmed that using three components was sufficient, and the loading plots showed which nutrients influenced each principal component the most. Overall, PCA proved useful in simplifying the data, uncovering hidden trends, and supporting clearer food classification and nutritional analysis.

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